

Spatial Urban Assessment & Decision-Support System

Pedestrian Access Screening around
Shahr-e Rey Square, Tehran

RESEARCH PROTOTYPE

Organising spatial evidence for expert review:
mapped services, modelled routes and visible uncertainty.

GIS • network analysis • evidence traceability • urban interpretation

A service nearby is only the starting point

Destination counts need a network and an evidence context.

01 / DESTINATIONS

What is mapped?

Services represented in the dataset.

Completeness of the real inventory remains unknown.

02 / NETWORK

How is it connected?

Model connectivity and route length.

A geometric radius does not cap network distance.

03 / EVIDENCE

What can we trust?

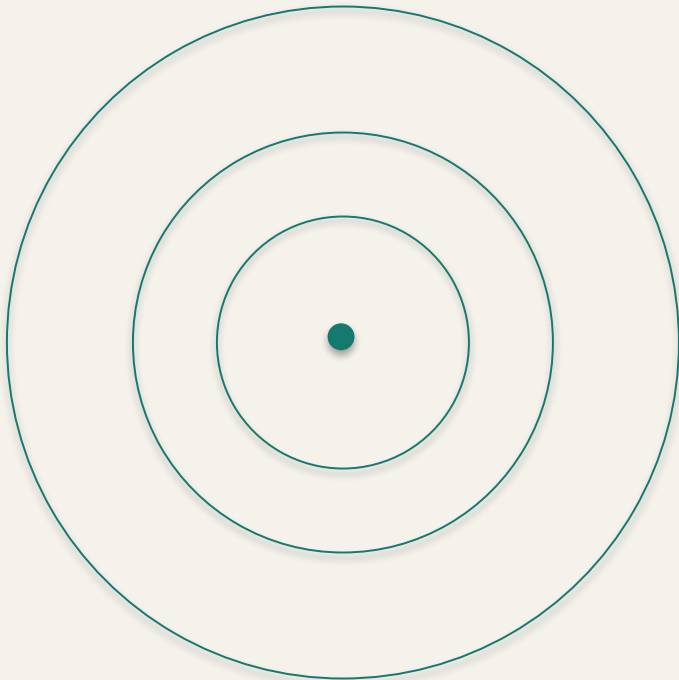
Provenance, missing tags and uncertainty.

Physical sidewalk quality needs observation.

Conceptual framework: destinations + network + evidence → reviewable urban questions

Geometric scales, network paths, evidence trail

Existing OpenStreetMap / Overpass outputs and modelled shortest network paths.



300 / 500 / 800 m

Conceptual buffers — not a map

1 Select mapped destinations

Representative coordinates within each geometric scale.

2 Calculate network connections and distances

Snapping to the extracted network; Dijkstra shortest paths.

3 Compare scales and inspect provenance

Main scope: 500 m. Overpass search extent: 800 m.

Geometric radius $\neq$ network distance $\neq$ physical route quality

Connectivity is not the same as route length

Reported values across geometric analysis scales.

DESTINATIONS & MODEL CONNECTIONS

Scale	Mapped	Connected	Ratio
300 m	59	59	100%
500 m	88	88	100%
800 m	177	164	92.7%

NETWORK DISTANCE (m) & MEAN DETOUR

Scale	Mean	Median	P90	Detour
300 m	254	210	506	1.67
500 m	392	288	829	1.64
800 m	610	650	1063	1.51

Connection ratio: connected / all mapped destinations in the same scale.

Distance summaries: connected destinations only (n = 59 / 88 / 164).

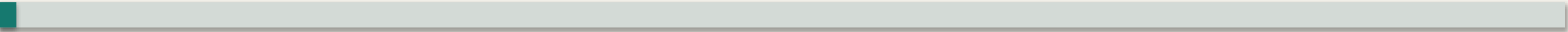
Network distance does not establish travel time, safety or physical sidewalk quality.

Missing sidewalk evidence is a central finding

Reported aggregate tagging profile by length — not physical sidewalk coverage.

98.9% unknown

1.1% known sidewalk-tag information



SCORE ≠ PHYSICAL QUALITY Legacy score 100 does not prove sidewalk quality.	13 NON-CONNECTIONS Cause unknown; not a confirmed network defect.	CONFIDENCE DISCREPANCY 0.87 vs 0.0 observed; cause and fix unproven.
UNKNOWN IN THE REAL WORLD Service completeness, route quality and square-centre alignment.	EMPTY OUTPUT ≠ NO ISSUE Empty diagnosis / recommendations do not establish absence of problems.	VALIDATION NOT ESTABLISHED No field validation; historical ACCESSIBILITY golden equivalence unproven.

Make the expert review easier to organise

Demonstrated capabilities: spatial analysis, network interpretation and traceable reporting.

01

FRAME

Question + analysis
scope

02

ASSEMBLE

Indicators +
available evidence

03

EXPOSE

Data gaps + review
flags

04

REVIEW

Human questions +
checks

Expert prompts: Is the origin correct? Is the service inventory sufficient?
Do modelled routes match actual entrances, crossings and sidewalk continuity?

Portfolio-ready with disclosed limitations

A limited field review can increase trust in this specific case.

PORTFOLIO READY WITH DISCLOSED LIMITATIONS

Review the origin → Shahr-e Rey taxi terminal

61 m

Reported direct distance

433 m

Reported network distance

FIELD REVIEW DELIVERABLES

Photos

Annotated map

Model / observation comparison

Proposed research check, not a prerequisite for current portfolio presentation.

Reported distances are not field-verified measurements.